

How to use the Green Fleet Planner

The planner answers one question: **which ships should run, on which fuel, at what speed — and what does each choice cost and emit?** This guide walks through every screen, every control and every number, in the order you will meet them.

1

Starting the planner

From the project folder, once:

```
uv pip install -e ".[dev,optim,api]"  
python scripts/train_model.py          # trains the fuel model (about 2 minutes)
```

Then every time:

```
python -m greenfleet.api                # then open http://127.0.0.1:8000
```

The browser opens on the **Plan** page and immediately runs the example fleet, so the first thing you see is a finished result (about 4 seconds). The status pill at the top right reads **ready** when the planner is idle, **searching** while it works. The bottom-left of the sidebar shows which model is loaded: *Model: 1,00,528 audited ship-years · 25,051 ships*.

If a yellow bar says "No trained model loaded", run `python scripts/train_model.py` and restart. The planner still works but predictions are physics-only estimates.

The sidebar has four pages: **Plan** (the main tool), **Fuels** (compare fuels over their whole life), **Calculator** (fuel burn for one ship at any speed), **Ports** (what each Indian port can supply, by year). Nothing needs the internet after the page has loaded.

Plan — the main screen

The setup card. Everything the planner needs is here; the results appear underneath as soon as you press **Find plans**.

Setting up

- 1 Fleet**
 Pick a starting fleet. *Coastal cargo fleet · 12 ships · 2047* is the demonstration case (clean fuels available at the ports). *2030* is the same fleet when only fossil fuels can be bought. *Mormugao harbour tugs* is the Green Tug Transition case. *Two ships · one route* is the simplest possible problem. Choosing a fleet also sets the year and the fuels; you can change them afterwards.
- 2 Year**
 Which year the plan is for. This decides which fuels the ports can actually supply (see **Ports**) and how clean India's electricity is. The planner will never assign a fuel that cannot be bought at every port on a route in that year.
- 3 Fuels you can buy**
 Tap a fuel to include or exclude it. A ticked chip is included. Six fuels: heavy fuel oil, diesel, LNG, methanol, ammonia, hydrogen. Include only what your ships could really run on and your ports could really supply.
- 4 Fleet line**
 A one-line summary of the fleet as it stands: number of ships, routes and ports, cargo to move each year. It says *edited* if you have changed ships or routes.
- 5 Carbon tax, ₹ per tonne CO₂**

A charge on every tonne of CO₂. Leave 0 if none. A tax makes dirty fuels cost more, so cleaner plans become the cheaper ones.

6 CO₂ limit, tonnes per year

Optional ceiling. Any plan above it is rejected. Leave blank for no limit.

7 Edit ships & routes

Opens a table where every cell is editable and rows can be added or removed. Column meanings are in the next section.

8 How each fuel is made

For fuels that can be produced in more than one way, choose the way. This matters enormously: ammonia made from natural gas ("grey") is **worse than diesel** over its whole life; made with renewable electricity ("green") it is far cleaner. Same for hydrogen. For methanol, choose *e-fuel* for the renewable version.

9 Population · Generations · Seed

How hard the search works. Population × Generations = number of complete plans evaluated (30 × 200 = 6,000, the same budget as the published benchmarks; a run takes 3–5 seconds). Raise Generations for a bigger fleet. Seed is the random starting point: the same seed always gives the same answer, a different seed explores differently.

10 Also run for comparison

Runs standard methods on the same problem with the same budget so you can see how our search compares: *NSGA-II* (the standard genetic algorithm), *Greedy* (what a planner does by hand), *Random*. They add a second or two.

11 Find plans

Starts the search. A progress bar shows plans checked and plans kept. The button is disabled while searching.

Editing ships and routes

COLUMN	MEANING
Name	Any label. Appears in the results.
Type	Ship class (bulk carrier, general cargo, container, tanker, harbour tug...). Sets the typical speed and hotel load.
Capacity t	Cargo the ship can carry per trip, tonnes.
EEDI/EIV	The energy-efficiency figure from the ship's certificate, gCO ₂ per tonne-mile. Leave blank if unknown; the planner then uses a typical value and marks the ship <i>estimate</i> .
Min kn · Max kn	Slowest and fastest speed the engine allows. The planner chooses a speed between them.
Tank m ³	Fuel tank volume. Decides whether a fuel has the range for the route (hydrogen needs a much bigger tank than diesel).
Cost ₹/yr	Yearly running cost if the ship is used: crew, maintenance, charter. Fuel is added on top by the planner.
Hours/yr	Hours the ship can be at sea per year after maintenance and port time.
Route: From · To	Ports at each end. Only the eight major ports in the Ports page are available.
Distance nmi	One-way distance in nautical miles (1 nmi = 1.852 km).
Cargo t/yr	Tonnes that must be moved on this route each year. Every plan shown delivers all of it.
Max hours	Optional deadline per one-way trip. Leave blank for none.

If the cargo is more than the fleet could move at full speed on every route, the planner refuses immediately and tells you the shortfall. Add ships, raise capacity, or lower the cargo figure.

Reading the results

Your options 18 plans deliver all 1.95 million t of cargo · 2047

● **Lowest cost**

₹69.5 cr **28,761 t**

per year CO₂ per year

+2% vs diesel · 3 of 12 ships

3 × Heavy fuel oil

X Misses Harit Sagar 2047

● **Balanced** Selected

₹85.3 cr **19,159 t**

per year CO₂ per year

-30% vs diesel · 3 of 12 ships

1 × Methanol 1 × LNG 1 × Heavy fuel oil

X Misses Harit Sagar 2047

● **Lowest emissions**

₹114 cr **2,602 t**

per year CO₂ per year

-91% vs diesel · 3 of 12 ships

3 × Methanol

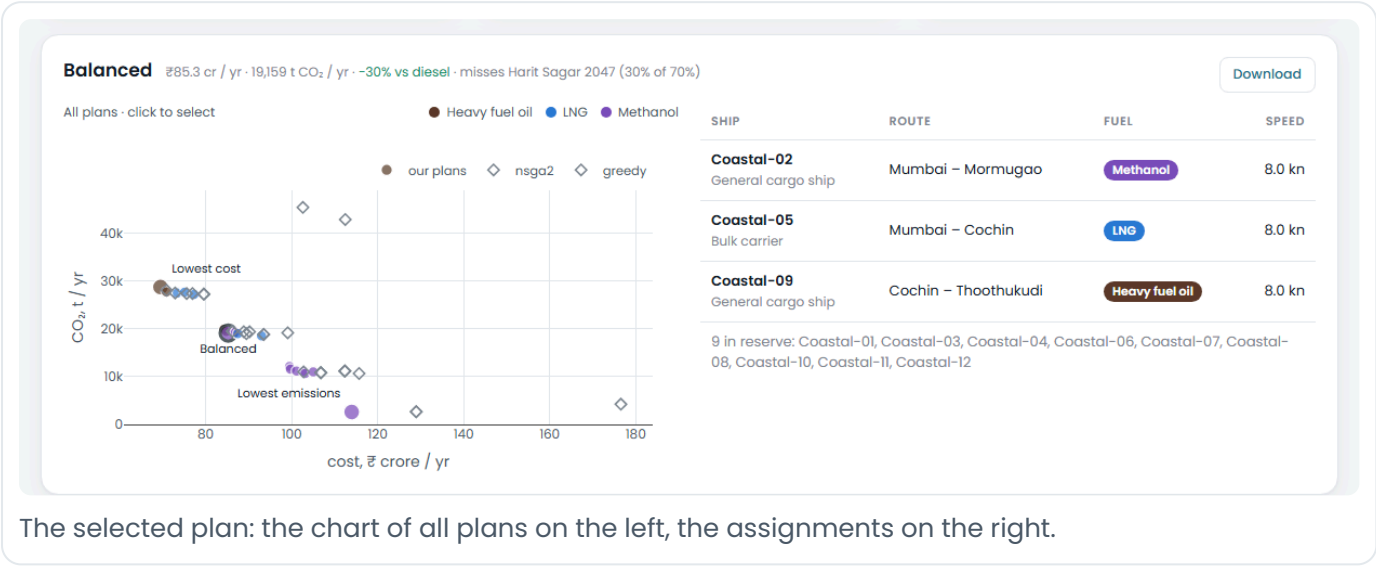
✓ Meets Harit Sagar 2047

Your options — three cards. Every plan shown delivers all the cargo; the difference is only cost and emissions.

- **Lowest cost** — the cheapest plan that delivers everything. Usually all fossil fuel.
- **Balanced** — the plan closest to "good on both" (selected by default).
- **Lowest emissions** — the cleanest plan that delivers everything.

On each card: **₹ per year** (fuel + running costs + any carbon tax), **tonnes of CO₂ per year** over the whole life of the fuel, **% vs diesel** (the same ships on the same routes burning diesel — green is less, red is more), **ships used**, the **fuel mix** as coloured chips, and whether the plan **meets or misses the Harit Sagar target** for that year (India's port target: 30% less CO₂ per tonne of cargo by 2030, 70% by 2047).

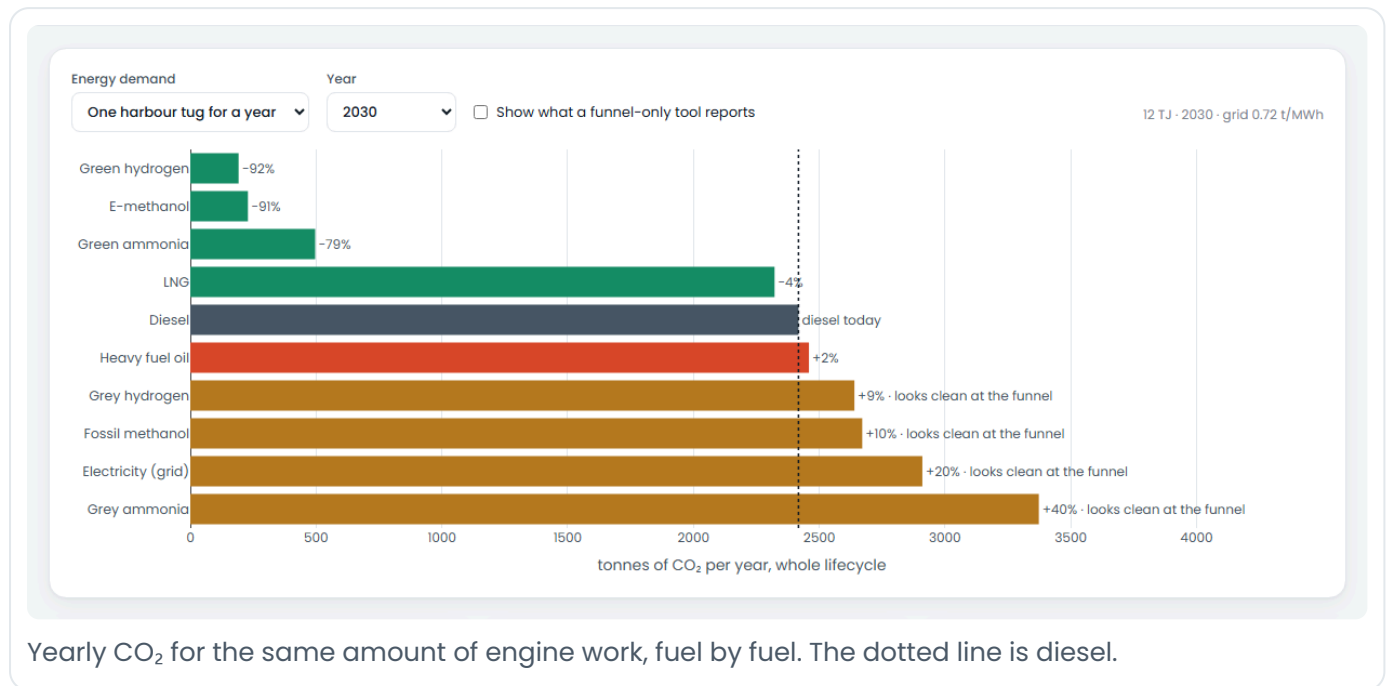
Click any card to select it.





PANEL	WHAT IT TELLS YOU
Evaluations	How many complete plans were scored, and how long it took.
Search space	How many combinations exist in total (ships × routes × fuels), and the number of Q-bits — the quantum-inspired decision variables — used to search it.
Hypervolume vs NSGA-II	Hypervolume measures how much of the cost–emissions space a set of plans covers; bigger is better. Above 1× means our search found a better set than the standard method on the same budget. On 12 ships this is roughly a tie (≈1.0–1.1×); the advantage grows with fleet size.
Repairs	Plans the search generated that broke a rule and were repaired rather than thrown away.
Search progress	Top: number of plans on the front per generation — it should climb. Bottom: Q-bit diversity from 1 (all options open) towards 0 (decided) — it should fall smoothly, not collapse.
Hypervolume vs baselines	The same score for each method side by side.
Provenance	Which model produced the fuel figures, the problem size, the fuel pathways and prices used, and a link that reopens exactly this result.

Fuels — compare fuels over their whole life

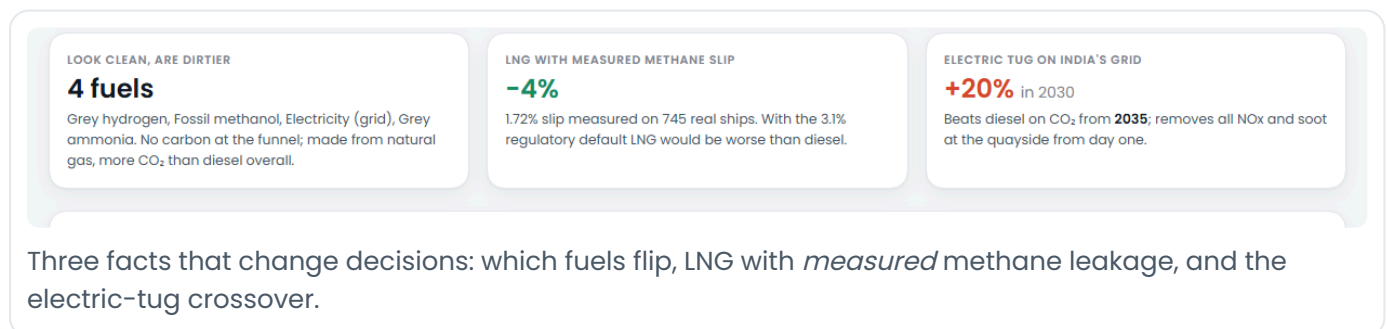


Energy demand — how much work the engines do in a year: one harbour tug, one coastal ship, or exactly 1 TJ for a per-unit comparison. Only the scale changes; the percentages do not.

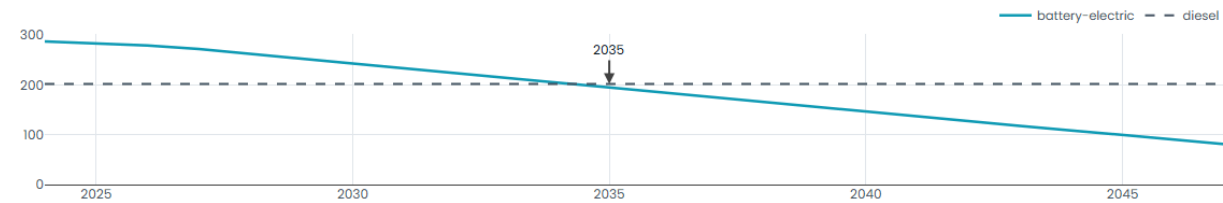
Year — matters only for electricity: India's grid gets cleaner each year, so electric options improve.

Show what a funnel-only tool reports — adds a black tick at the figure a tool would give if it counted only what leaves the funnel. For ammonia, hydrogen and electricity that tick sits at zero; the bar shows the truth.

Colours: green bars are cleaner than diesel, red are dirtier, **amber means "looks clean at the funnel but is dirtier over the whole life"** — those are the fuels a naive comparison gets wrong.



Electric tug vs diesel, by year



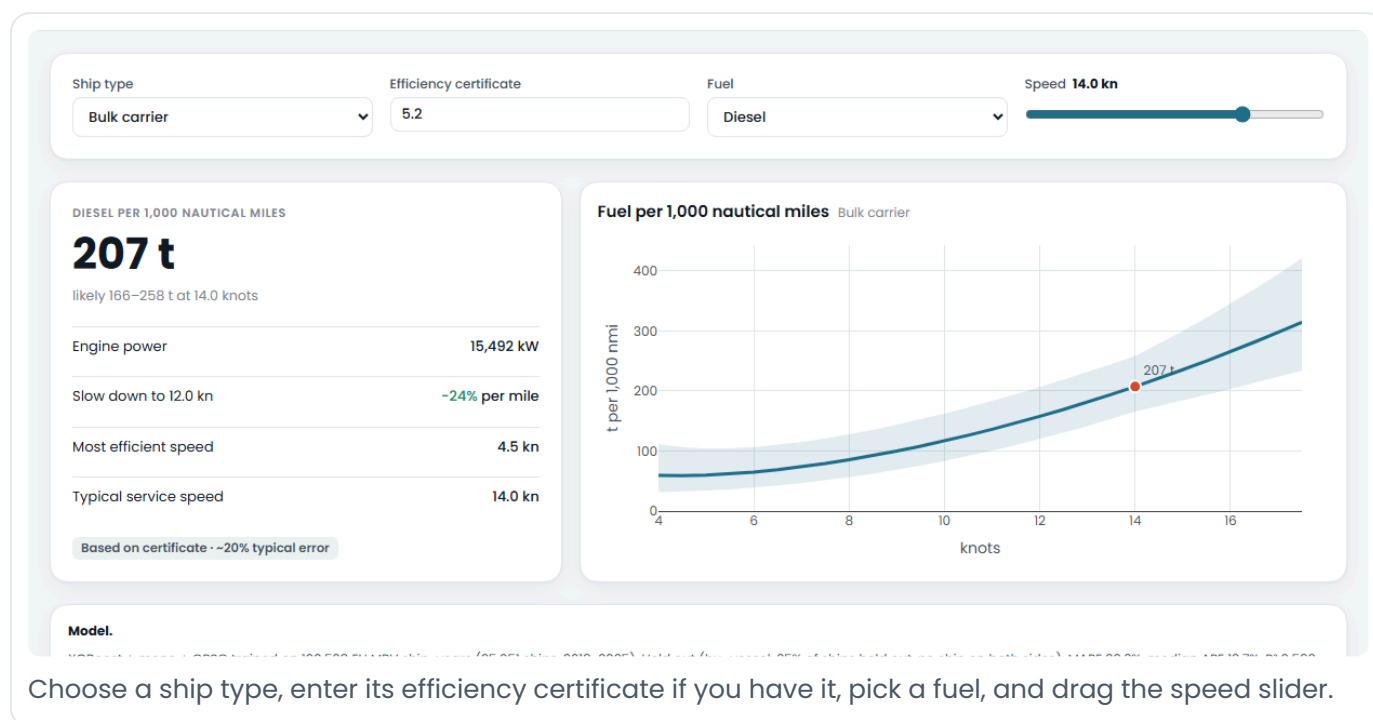
Lifecycle breakdown, tonnes CO₂e per year

OPTION	TOTAL T	UPSTREAM	CO ₂	CH ₄	N ₂ O	PILOT	VS DIESEL	FUNNEL-ONLY	CONFIDENCE
Diesel	2,418	384	2,002	1	31	0	0%	0%	medium
Heavy fuel oil	2,459	360	2,066	1	33	0	+2%	+3%	medium
LNG	2,323	493	1,528	284	17	0	-4%	-24%	medium
Fossil methanol	2,671	801	1,769	0	4	97	+10%	-12%	provisional
E-methanol	229	128	0	0	4	97	-91%	-100%	provisional
Grey ammonia	3,373	3,001	0	0	203	169	+40%	-100%	provisional
Green ammonia	496	124	0	0	203	169	-79%	-100%	provisional
Grey hydrogen	2,640	2,640	0	0	0	0	+9%	-100%	provisional
Green hydrogen	192	192	0	0	0	0	-92%	-100%	provisional
Electricity (grid)	2,910	2,910	0	0	0	0	+20%	-100%	provisional

Electric tug vs diesel by year (the arrow marks the crossover year), and the full lifecycle breakdown.

COLUMN	MEANING
Total t	Tonnes of CO ₂ -equivalent per year over the whole life of the fuel.
Upstream	Making and transporting the fuel before it reaches the ship.
CO ₂	From burning it on board.
CH ₄	Methane that escapes unburnt (LNG), converted to CO ₂ -equivalent.
N ₂ O	Nitrous oxide formed in the engine (mostly ammonia), converted to CO ₂ -equivalent.
Pilot	The small amount of diesel a methanol or ammonia engine needs to ignite.
vs diesel · Funnel-only	Change against diesel over the whole life, and what a funnel-only count would claim.
Confidence	How well established the underlying factor is. <i>Provisional</i> figures should be confirmed before publication.

Calculator — fuel for one ship at any speed



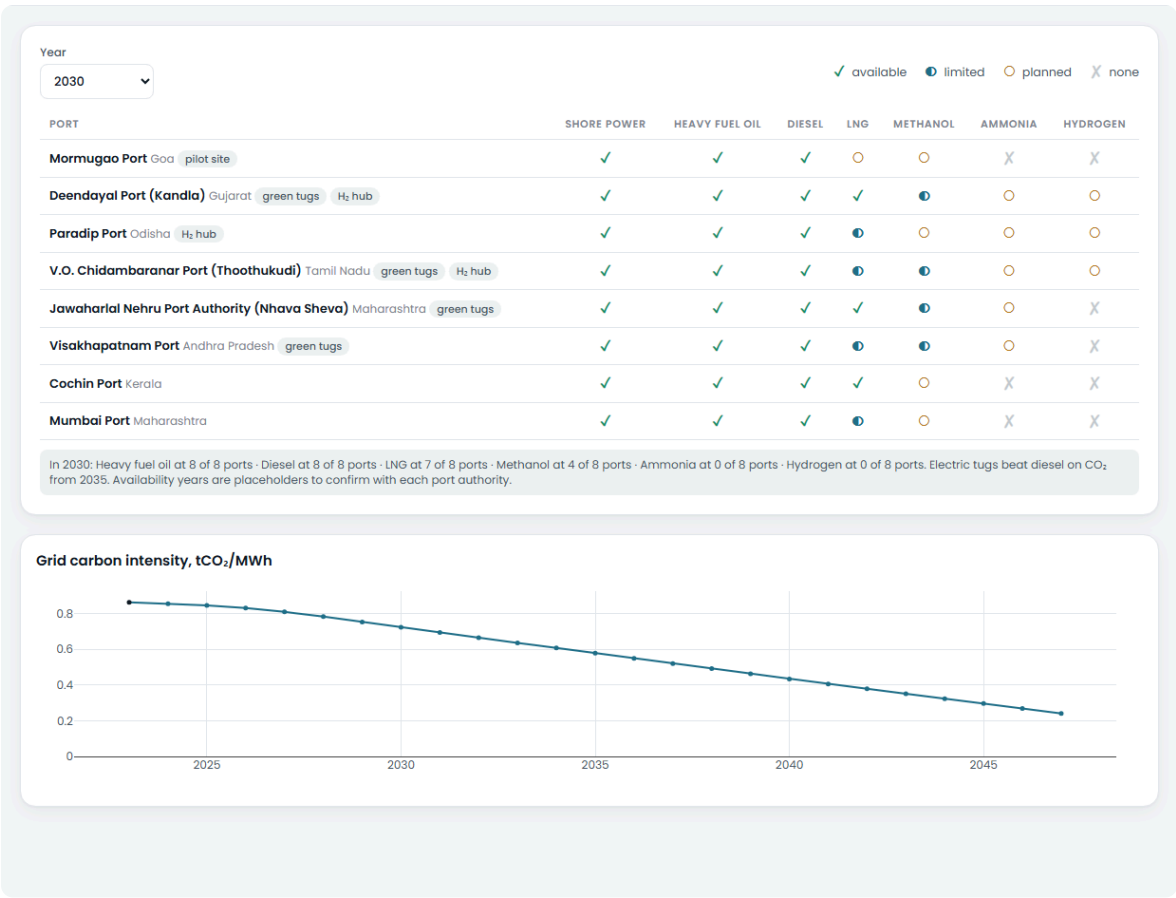
The big number is **tonnes of fuel per 1,000 nautical miles** at the chosen speed, with the likely range. Below it: engine power needed, how much slowing down by two knots saves per mile, the most fuel-efficient speed if there were no schedule, and the typical service speed for that ship type.

The chart is the same figure at every speed: the line is the estimate, the shaded band the likely range, the red dot your chosen speed. Slower is cheaper per mile until very slow speeds, where the ship's hotel load (lights, cooling, pumps) starts to dominate.

The badge says what the figure is based on. **Based on certificate** means the ship's own efficiency figure was used and the model is typically within about 20% on ships it has never seen. **Estimate** means no certificate was given — the planner uses a typical value for the type and shows a much wider range; plan with the upper end. Harbour tugs are always estimates: they are below the size the data covers.

5

Ports — what each port can supply



Eight major Indian ports; shore power and each fuel, for the chosen year.

✓ available · ● limited (small volumes, by arrangement) · ○ planned, **not usable yet** · ✕ none. The Plan page uses this table: a fuel is allowed on a route only if every port on the route shows ✓ or ● in that year. "Planned" is deliberately not enough — a fleet plan built on an uncommissioned facility strands ships.

Chips mark the *pilot site* (Mormugao), ports in the *green tugs* programme, and *H₂ hub* ports designated for green hydrogen.

The grid chart shows the carbon intensity of India's electricity used for shore power and battery-electric ships; values after 2023 are projections.

The years in this table are placeholders with the correct structure. Before any published result, confirm them with each port authority — this is the first task of the pilot.

Messages you may see, and what to do

MESSAGE	MEANING	WHAT TO DO
Select at least one fuel.	No fuel chip is ticked.	Tick at least one.
This fleet cannot move that much cargo...	Even at full speed on every route the ships cannot deliver the cargo figure.	Add ships, raise capacity or hours, or lower the cargo.
No plan delivers all the cargo with these fuels in this year...	A ticked fuel cannot be bought at one of the ports on a route in that year, or the CO ₂ limit is too tight.	Check the Ports page for the year; tick diesel as well; raise or remove the CO ₂ limit.
Yellow bar: Planning service unreachable	The server is not running.	Start it with <code>python -m greenfleet.api</code> and reload.
Yellow bar: No trained model loaded	The fuel model has not been trained on this machine.	Run <code>python scripts/train_model.py</code> , restart the server.
A ship marked <i>estimate</i>	No efficiency certificate was given for it.	Enter the EEDI/EIV in Edit ships & routes if you have it.

A five-minute walkthrough

1 Open on the finished result

Read the three cards left to right: cheapest (all HFO, misses the target) → balanced → cleanest (all e-methanol, 91% less than diesel, meets Harit Sagar 2047). Same ships, same cargo.

2 Click Lowest emissions

Show the assignments: which ship, which route, which fuel, what speed, who stays in reserve. Click a middle dot on the chart to show there are more plans than three.

3 Switch the fleet to 2030 and press Find plans

Every plan is fossil, every card misses the target. Point at Ports: in 2030 methanol is only *planned* at Mormugao and Cochin, so the planner refuses to use it.

4 Untick every fuel except Hydrogen, press Find plans

"No plan delivers all the cargo" — hydrogen is not bunkerable on these routes in that year. The refusal is the feature.

5 Fuels page

Tick "Show what a funnel-only tool reports": the black ticks at zero for ammonia and hydrogen against the amber bars. Then the three fact cards.

6 Calculator

Drag the speed from 14 to 12 knots: -24% per mile. Switch the type to Harbour tug: the badge turns to *Estimate* and the band widens — the tool says when it is guessing.

7 Back to Plan, scroll to How the search performed

6,000 plans evaluated in a few seconds out of 2.3×10^{13} combinations; hypervolume about level with NSGA-II on 12 ships; the diversity curve falling smoothly.

Companion documents: [docs/STUDY_GUIDE.pdf](#) (what the project is and why, in plain language) · [docs/REPORT_FOR_PPT.pdf](#) (results for the deck) · [docs/DEMO_GUIDE.md](#) (what to say at each screen) · [docs/FINDINGS_PHASE4.md](#) (how the platform is built).